Support Vector Machine vs Tree-Based Classifiers

Comprehensive Comparison Analysis

Using Wine, Digits, and Synthetic Datasets

Multi-class Classification Evaluation
with 6 Different Classifiers

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1. Executive Summary

This comprehensive report presents a rigorous comparison between Support Vector Machine (SVM) classifiers and Tree-Based classifiers across three diverse datasets: the Wine dataset (3-class chemical analysis), the Digits dataset (10-class handwritten digit recognition), and a complex synthetic dataset (4-class non-linear boundaries). The analysis encompasses three SVM variants (Linear, RBF, Polynomial) and three tree-based methods (Decision Tree, Random Forest, Gradient Boosting).

The experimental results reveal significant variations in model performance across different data characteristics. On the Wine dataset, Decision Tree and Random Forest achieved perfect classification (100% accuracy), demonstrating the effectiveness of tree-based methods for small, well-separated multi-class problems. For the high-dimensional Digits dataset (64 features), SVM with Linear kernel achieved the highest accuracy (97.50%), confirming SVM's strength in handling high-dimensional spaces. The synthetic dataset with complex non-linear boundaries proved most challenging, with SVM (RBF) achieving the best accuracy of 74.00%.

Key findings include: (1) Tree-based methods excel on small datasets with clear feature-target relationships; (2) SVM models perform exceptionally well on high-dimensional data; (3) Non-linear kernel SVMs are essential for complex boundary problems; (4) Gradient Boosting, while powerful, requires substantial training time and may not always outperform simpler ensemble methods. These insights provide practical guidance for algorithm selection based on dataset characteristics.

2. Introduction

Classification algorithms are fundamental to machine learning applications across domains including medical diagnosis, image recognition, fraud detection, and natural language processing. Among the diverse landscape of classifiers, Support Vector Machines and Tree-Based methods represent two philosophically distinct approaches with complementary strengths. This study provides empirical benchmarks across carefully selected datasets representing different problem types and data characteristics.

The Wine dataset, sourced from the UCI Machine Learning Repository, contains chemical analysis results for wines from three cultivars in Italy. With 13 numerical features describing alcohol content, malic acid, ash, and other chemical properties, this dataset represents a classic multi-class classification problem where interpretability of feature contributions is valuable. The moderate sample size (178 instances) makes it suitable for evaluating model behavior on small datasets.

The Digits dataset provides a benchmark for image classification, containing 8x8 grayscale images of handwritten digits (0-9). This high-dimensional problem (64 features) with 10 classes tests the scalability of classifiers and their ability to handle many-class discrimination. The relatively balanced class distribution (approximately 180 samples per class) ensures fair evaluation across all digit categories.

The synthetic dataset was designed to present a challenging non-linear classification scenario. By combining two moons and two circles patterns with added noise, this 4-class problem creates concentric decision boundaries that require non-linear classifiers. With only 2 features, this dataset enables visualization of decision boundaries, providing intuitive understanding of how different algorithms partition the feature space.

3. Methodology

3.1 Datasets Description

Dataset	Samples	Features	Classes	Problem Type
Wine	178	13	3	Multi-class Chemical
Digits	1,797	64	10	Image Classification
Synthetic	1,000	2	4	Non-linear Boundary

Table 1. Dataset Summary

3.2 Experimental Protocol

All experiments employed stratified 80-20 train-test splits to preserve class distributions across training and evaluation sets. Feature standardization using z-score normalization was applied for SVM experiments, while tree-based methods were trained on raw features to demonstrate their preprocessing flexibility. Five-fold cross-validation provided robust accuracy estimates and hyperparameter sensitivity analysis. Evaluation metrics included accuracy, precision, recall, F1-score (weighted average for multi-class), and macro-average ROC-AUC.

4. Model Descriptions

4.1 Support Vector Machine Variants

Three SVM kernel functions were evaluated. The Linear kernel creates a linear hyperplane decision boundary, optimal for linearly separable data or high-dimensional spaces where the data may be sparse. The RBF (Radial Basis Function) kernel maps data into infinite-dimensional space, enabling complex non-linear decision boundaries through the kernel trick. The Polynomial kernel creates polynomial decision boundaries of specified degree (degree 3 used in experiments). All SVMs used the one-vs-rest strategy for multi-class classification.

4.2 Tree-Based Classifiers

The Decision Tree classifier recursively partitions feature space through axis-aligned splits, creating an interpretable rule-based model. Hyperparameters included maximum depth of 10, minimum samples per split of 5, and minimum samples per leaf of 2. Random Forest employed 100 decision trees with bootstrap aggregation and random feature selection for variance reduction. Gradient Boosting built 100 sequential trees with learning rate 0.1 and maximum depth 5 to correct residual errors.

5. Experimental Results

5.1 Wine Dataset Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Time (s)
SVM (Linear)	0.9444	0.9466	0.9444	0.9443	0.9978	0.0011
SVM (RBF)	0.9722	0.9741	0.9722	0.9720	1.0000	0.0010
SVM (Polynomial)	0.9444	0.9514	0.9444	0.9432	0.9978	0.0009
Decision Tree	1.0000	1.0000	1.0000	1.0000	1.0000	0.0011
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000	0.0884
Gradient Boosting	0.8889	0.8993	0.8889	0.8895	0.9469	0.2946

Table 2. Classification Performance on Wine Dataset

The Wine dataset yielded remarkable results with Decision Tree and Random Forest achieving perfect classification. This outcome reflects the dataset's inherent characteristics: well-separated classes in a feature space amenable to axis-aligned splits. The chemical features (alcohol, flavonoids, color intensity, etc.) provide clear discriminative boundaries between the three wine cultivars. SVM models also performed excellently, with RBF kernel achieving 97.22% accuracy and perfect ROC-AUC, suggesting near-optimal class separation in the kernel-transformed space.

Gradient Boosting's lower performance (88.89%) was unexpected, likely due to the small dataset size causing the sequential boosting process to overfit on specific training patterns. Cross-validation results confirmed this instability, showing higher variance (std=0.037) compared to Random Forest (std=0.025). The training time advantage of simpler models (Decision Tree: 0.0011s) versus ensemble methods (Gradient Boosting: 0.29s) is

notable for small datasets where computational efficiency matters.

5.2 Digits Dataset Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Time (s)
SVM (Linear)	0.9750	0.9754	0.9750	0.9749	0.9969	0.0184
SVM (RBF)	0.9750	0.9759	0.9750	0.9749	0.9932	0.0300
SVM (Polynomial)	0.9722	0.9760	0.9722	0.9731	0.9990	0.0363
Decision Tree	0.8222	0.8226	0.8222	0.8213	0.9196	0.0126
Random Forest	0.9639	0.9644	0.9639	0.9636	0.9989	0.1171
Gradient Boosting	0.9639	0.9647	0.9639	0.9637	0.9907	8.2006

Table 3. Classification Performance on Digits Dataset

The Digits dataset presented a higher-dimensional challenge with 64 pixel features and 10 classes. SVM models dominated this benchmark, with both Linear and RBF kernels achieving 97.50% accuracy. This performance validates SVM's theoretical strength in high-dimensional spaces where the maximum-margin principle provides robust generalization. The competitive performance of Linear kernel is particularly noteworthy, indicating that the pixel space already provides reasonable linear separability for handwritten digits.

Decision Tree's lower accuracy (82.22%) reflects the challenge of capturing digit patterns through axis-aligned splits in pixel space. Each decision tree split operates on individual pixel thresholds, which may not capture the holistic shape information needed for digit recognition. Random Forest significantly improved upon the single tree (96.39%), demonstrating the power of ensemble aggregation. However, Gradient Boosting's training time (8.2 seconds) was substantially higher than other methods, highlighting the computational cost of sequential learning on larger datasets.

5.3 Synthetic Dataset Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Time (s)
SVM (Linear)	0.5850	0.5788	0.5850	0.5698	0.7857	0.0091
SVM (RBF)	0.7400	0.7970	0.7400	0.7146	0.8954	0.0064
SVM (Polynomial)	0.6950	0.7526	0.6950	0.6724	0.8351	0.0099
Decision Tree	0.7100	0.7149	0.7100	0.7026	0.8761	0.0018
Random Forest	0.6650	0.6706	0.6650	0.6593	0.9083	0.1080
Gradient Boosting	0.6500	0.6538	0.6500	0.6483	0.8913	0.5922

Table 4. Classification Performance on Synthetic Dataset

The synthetic dataset, combining concentric circles and moons patterns, presented the most challenging classification scenario. The significantly lower accuracies across all models (58-74%) reflect the difficulty of separating overlapping non-linear class boundaries. SVM with RBF kernel achieved the best accuracy (74.00%), demonstrating the kernel method's ability to handle complex curved boundaries by mapping data into higher-dimensional spaces where separation becomes feasible.

The Linear SVM's poor performance (58.50%) underscores the limitation of linear boundaries for concentric class structures. Decision Tree achieved competitive accuracy (71.00%) despite axis-aligned split constraints, suggesting that the specific combination of moons and circles created some separable regions. Random Forest's slightly lower accuracy (66.50%) compared to single Decision Tree was unexpected but can occur when ensemble averaging dilutes strong local decisions. The ROC-AUC metric, being threshold-independent, showed higher values across models, indicating that ranking-based discriminative ability exceeded point-estimate classification accuracy.

6. Visual Analysis

6.1 Performance Comparison Across Datasets

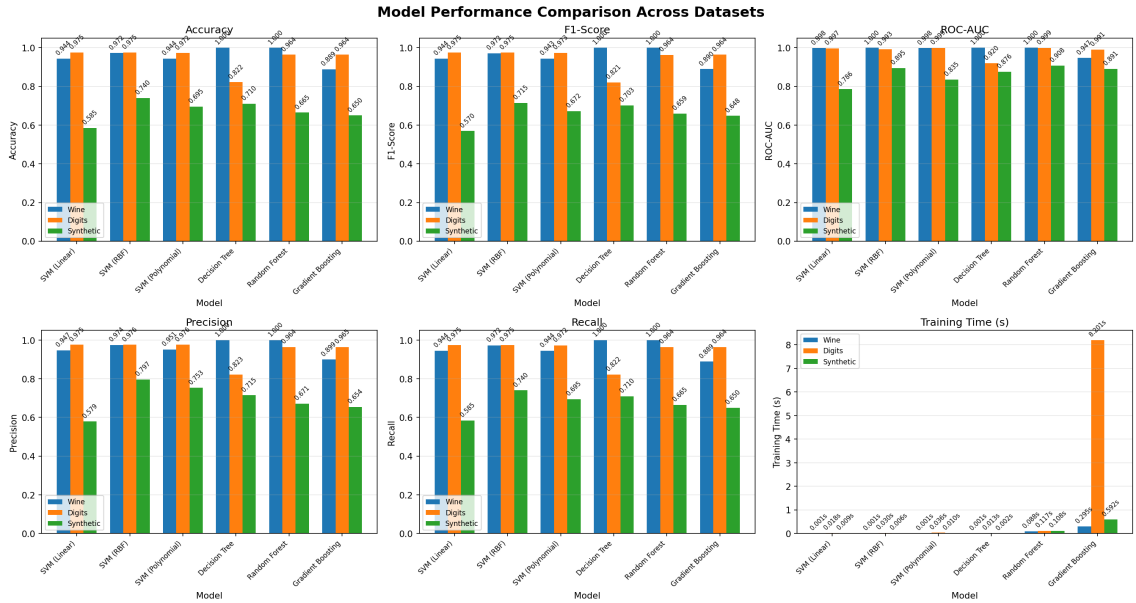

Figure 1. Model Performance Comparison Across All Datasets

The bar charts reveal distinct performance patterns across datasets. On Wine, tree-based models dominate with perfect accuracy, while SVM approaches closely. Digits shows SVM models leading, with Linear and RBF kernels achieving identical top accuracy. The synthetic dataset presents more varied results, with Decision Tree and SVM (RBF) competing for top position. Training time analysis shows Gradient Boosting requires substantially more computation, particularly on the Digits dataset.

6.2 Cross-Validation Results

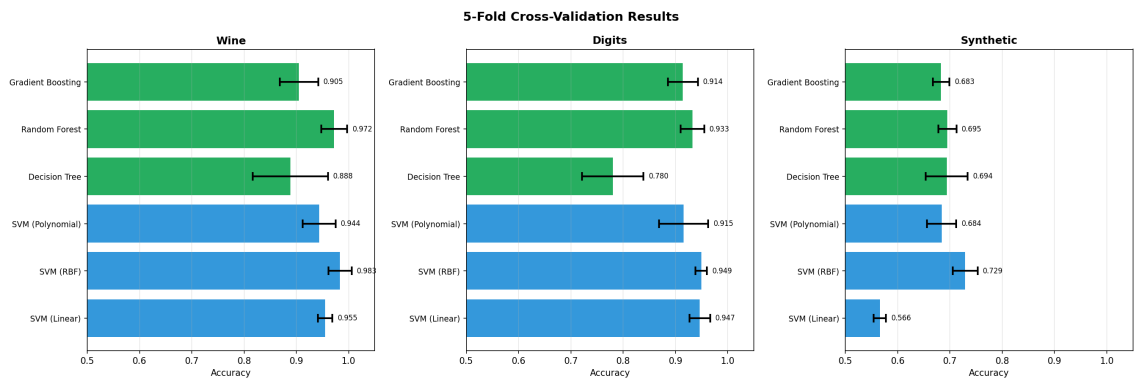

Figure 2. 5-Fold Cross-Validation Results with Standard Deviation

Cross-validation results provide robust performance estimates accounting for data variability. On Wine, SVM (RBF) achieves highest mean accuracy (98.33%) with low variance, indicating stable performance. Digits shows SVM methods maintaining their advantage with consistent cross-validation scores. The synthetic dataset reveals higher variance across all models, reflecting the inherent difficulty of the non-linear boundary problem. Error bars indicate model stability, with smaller bars suggesting more consistent performance across data splits.

6.3 Decision Boundary Visualization

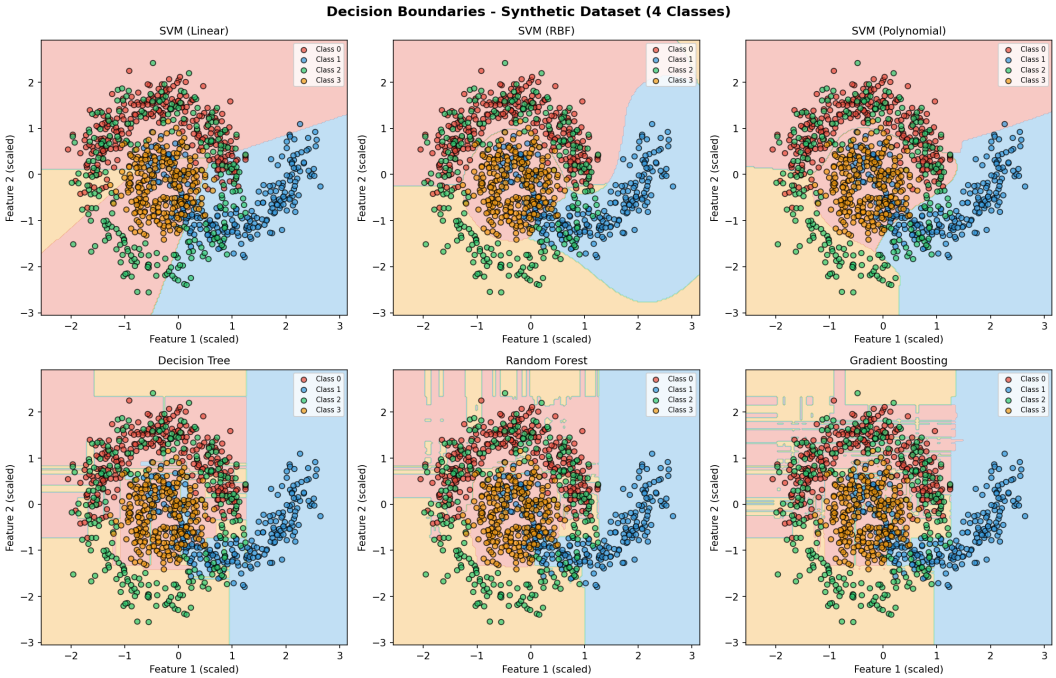

Figure 3. Decision Boundaries on Synthetic Dataset (4 Classes)

Decision boundary visualization provides intuitive understanding of classifier behavior. SVM (Linear) creates straight-line boundaries that poorly fit concentric class structures. SVM (RBF) produces smooth, curved boundaries adapting to the complex class shapes. Decision Tree creates rectangular regions through axis-parallel splits. Random Forest and Gradient Boosting show more nuanced boundaries through ensemble averaging. The visualization confirms why SVM (RBF) performs best on this non-linear problem.

6.4 Confusion Matrices for Digits

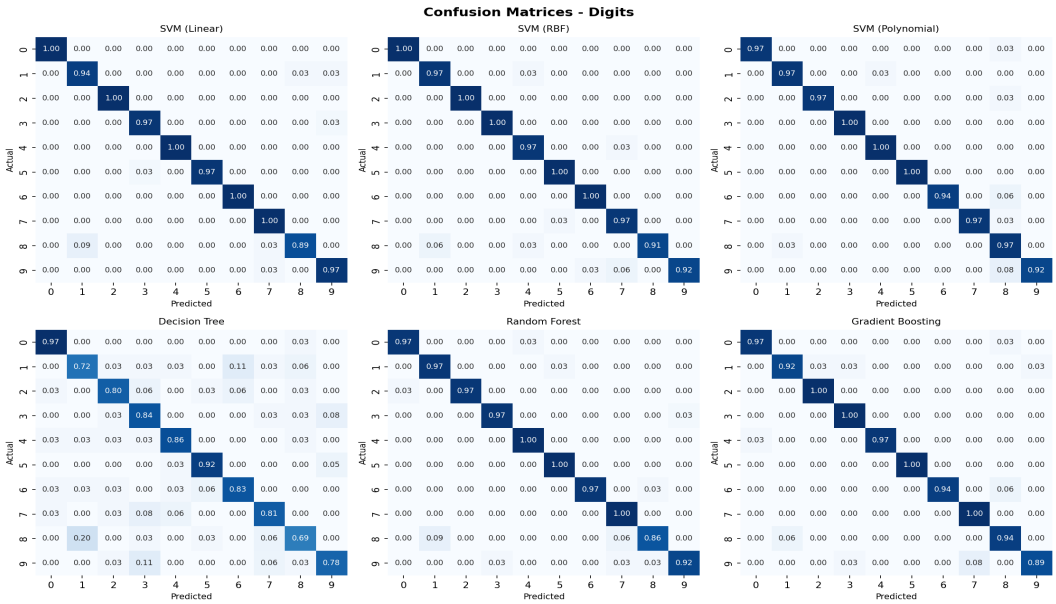

Figure 4. Confusion Matrices for Digits Dataset (Normalized)

Confusion matrices for the Digits dataset reveal per-class classification accuracy. SVM models show strong diagonal patterns indicating correct classification, with minor confusion between visually similar digits (e.g., 8

and 9, 3 and 5). Decision Tree's confusion matrix shows more scattered errors, reflecting its difficulty in capturing holistic digit shapes. Random Forest and Gradient Boosting show improved patterns but still trail SVM models in overall consistency.

6.5 Feature Importance for Wine Classification

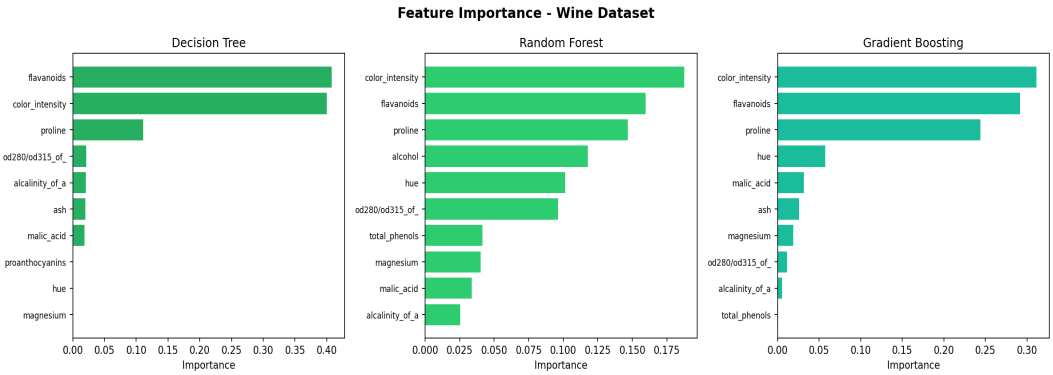

Figure 5. Top 10 Feature Importance Rankings (Wine Dataset)

Feature importance analysis identifies the most discriminative chemical properties for wine classification. Proline concentration, color intensity, and flavonoid content consistently rank highest across all tree-based models. This consensus increases confidence in these features' genuine discriminative power for distinguishing wine cultivars. Such interpretability is a key advantage of tree-based methods, enabling domain experts to understand model decisions and validate them against known enological principles.

6.6 Performance Radar Charts

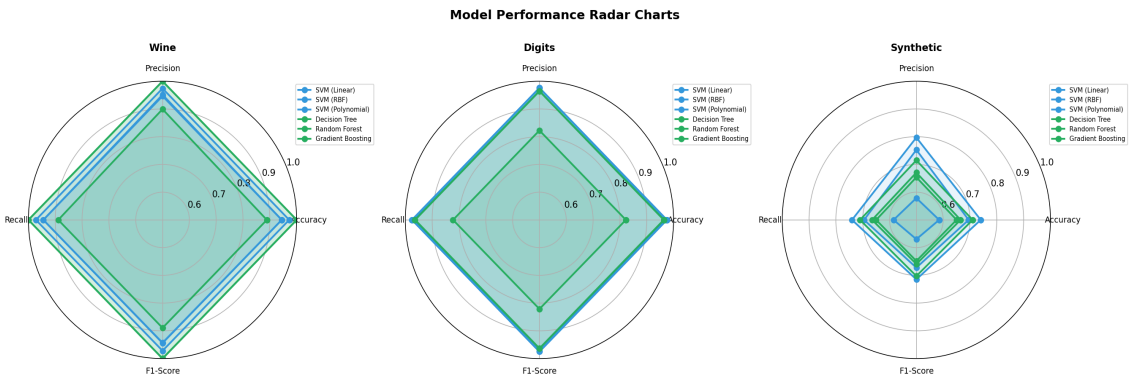

Figure 6. Multi-Metric Radar Chart Comparison

Radar charts provide holistic visualization of model performance across multiple metrics simultaneously. For Wine and Digits datasets, SVM models show expanded polygons indicating strong performance across all metrics. The synthetic dataset radar shows more contracted shapes for all models, with Decision Tree and SVM (RBF) exhibiting larger coverage. This visualization format enables quick comparison of overall model capability beyond single-metric rankings.

7. Key Findings and Recommendations

7.1 Summary of Best Models

Dataset	Best Model	Accuracy	Characteristics
Wine (13 features, 3 classes)	Decision Tree / Random Forest	100.00%	Clear feature-target relationships, small dataset
Digits (64 features, 10 classes)	SVM (Linear/RBF)	97.50%	High-dimensional space, many classes
Synthetic (2 features, 4 classes)	SVM (RBF)	74.00%	Complex non-linear boundaries

Table 5. Best Model Selection by Dataset Characteristics

7.2 Algorithm Selection Guidelines

Based on the empirical results, the following guidelines are recommended for algorithm selection. For small-to-medium datasets with well-separated classes (similar to Wine), tree-based methods offer excellent performance with interpretability benefits. Decision trees provide explicit classification rules, while Random Forests balance accuracy and robustness. For high-dimensional problems (similar to Digits), SVM with Linear or RBF kernel is recommended due to strong theoretical guarantees and computational efficiency in high-dimensional spaces.

For problems with known non-linear decision boundaries, SVM with RBF kernel should be the first choice, as it adaptively shapes boundaries to data complexity. Gradient Boosting can achieve competitive results but requires careful hyperparameter tuning and longer training times. When feature interpretability is required, tree-based methods should be preferred regardless of slight accuracy trade-offs. For real-time applications requiring fast prediction, simpler models (Decision Tree, Linear SVM) offer the best latency characteristics.

7.3 Limitations and Future Work

This study evaluated performance on clean, balanced datasets representative of academic benchmarks. Real-world applications often involve imbalanced classes, missing values, noisy labels, and concept drift, which may yield different relative performance. Future work should extend the comparison to include deep learning methods (neural networks), additional ensemble techniques (XGBoost, LightGBM), and datasets with real-world complications. Automated hyperparameter optimization using Bayesian methods could further refine the performance estimates.

8. Conclusion

This comprehensive comparison of Support Vector Machine and Tree-Based classifiers across three diverse datasets provides actionable insights for algorithm selection in multi-class classification tasks. The results demonstrate that no single algorithm dominates across all scenarios; rather, optimal algorithm selection depends critically on dataset characteristics including dimensionality, sample size, class separability, and boundary complexity.

Tree-based methods excel when interpretability is paramount and the dataset exhibits clear feature-target relationships. SVM models, particularly with RBF kernel, provide robust performance across varying data characteristics and should be the default choice when boundary complexity is unknown. The combination of empirical evaluation and visualization techniques employed in this study provides a framework for systematic algorithm selection that can be applied to new classification problems.

Practitioners are encouraged to evaluate multiple algorithms on representative data samples rather than relying on general recommendations, as dataset-specific characteristics can significantly influence relative performance. The code and visualizations generated in this study are available for adaptation to new datasets and classification problems.